Chapter 1: Relations and Functions

Practice Set A: step-wise solutions

20 questions **52** marks **3** levels

Each row is one line of working. The red number in the right margin is what an examiner gives for that line, as in the CBSE marking scheme (½ is half a mark). Write every line that carries a mark. ✓ ½

Level 1

8 marks

1 **MCQ** CBSE 2024 (65/1/3) Q7 — 1 mark

A relation R defined on a set of human beings as $R = \{(x, y) : x \text{ is 5 cm shorter than } y\}$ is:

(A) reflexive only

(B) reflexive and transitive

(C) symmetric and transitive

(D) **neither transitive, nor symmetric, nor reflexive**

Formula / idea: Reflexive: $(a, a) \in R$; symmetric: $(a, b) \in R \Rightarrow (b, a) \in R$; transitive: $(a, b), (b, c) \in R \Rightarrow (a, c) \in R$.

1 Reflexive? Nobody is 5 cm shorter than himself.
$(x, x) \notin R$

2 Symmetric? If x is 5 cm shorter than y, then y is 5 cm taller.
$(x, y) \in R \Rightarrow (y, x) \notin R$

3 Transitive? Let x be 5 cm shorter than y, and y than z.
x is 10 cm shorter than $z \Rightarrow (x, z) \notin R$ ✓ 1

4 R has none of the three properties. Option (D).

Answer **(D) neither transitive, nor symmetric, nor reflexive**

Watch out: Treating it like ‘shorter than’, which is transitive — ‘exactly 5 cm shorter’ is not.

2 MCQ CBSE 2024 (65/1/2) Q13 1 mark

A relation R defined on set $A = \{x : x \in \mathbb{Z} \text{ and } 0 \le x \le 10\}$ as $R = \{(x, y) : x = y\}$ is given to be an equivalence relation. The number of equivalence classes is:

(A) 1 **(B)** 2 **(C)** 10 **(D)** 11

Formula / idea: The equivalence class $[a]$ is the set of all elements related to a.

1 List the elements of A.
$A = \{0, 1, 2, \ldots, 10\} \Rightarrow n(A) = 11$

2 Each element is related only to itself.
$[a] = \{a\}$ for every $a \in A$

3 So there is one class for each element. ✓ 1
Number of equivalence classes $= n(A) = 11$

4 Option (D).

Answer **(D)** 11

Watch out: Counting 0 to 10 as ten numbers — there are eleven.

3 MCQ CBSE 2024 (65/3/1) Q2 1 mark

Let $\mathbb{R}_+$ denote the set of all non-negative real numbers. Then the function $f : \mathbb{R}_+ \to \mathbb{R}_+$ defined as $f(x) = x^2 + 1$ is:

(A) **one-one but not onto** **(B)** onto but not one-one
(C) both one-one and onto **(D)** neither one-one nor onto

Formula / idea: One-one: $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$. Onto: every element of the codomain has a pre-image.

1 One-one: let $f(x_1) = f(x_2)$ with $x_1, x_2 \ge 0$.
$x_1^2 + 1 = x_2^2 + 1 \Rightarrow x_1^2 = x_2^2 \Rightarrow x_1 = x_2$
both are non-negative, so $x_1 = -x_2$ is ruled out

2 Find the least value of f on $\mathbb{R}_+$.
$x^2 \ge 0 \Rightarrow f(x) \ge 1$

3 So 0 in the codomain has no pre-image: not onto. ✓ 1
$f(x) = 0 \Rightarrow x^2 = -1$, no real x

4 One-one but not onto. Option (A).

Answer **(A) one-one but not onto**

Watch out: Calling it many-one as for x^2 on $\mathbb{R}$ — this domain has no negative numbers.

4 MCQ CBSE 2024 (65/1/1) Q1 1 mark

A function $f : \mathbb{R}_+ \to \mathbb{R}$ (where $\mathbb{R}_+$ is the set of all non-negative real numbers) defined by $f(x) = 4x + 3$ is:

(A) both one-one and onto **(B)** onto but not one-one

(C) **one-one but not onto** **(D)** neither one-one nor onto

Formula / idea: Onto means range = codomain; here the codomain is all of $\mathbb{R}$.

1 One-one: let $f(x_1) = f(x_2)$.
$4x_1 + 3 = 4x_2 + 3 \Rightarrow x_1 = x_2$

2 Onto? Take $y = -1$ in the codomain $\mathbb{R}$.
$4x + 3 = -1 \Rightarrow x = -1 \notin \mathbb{R}_+$ ✓ 1

3 So -1 has no pre-image: f is not onto.

4 One-one but not onto. Option (C).

Answer **(C) one-one but not onto**

Watch out: Forgetting the domain is only $x \geq 0$; on all of $\mathbb{R}$ the same rule is onto.

5 MCQ Board-pattern 1 mark

Let $A = \{1, 2, 3, 4\}$ and $B = \{a, b\}$. The number of onto functions from A to B is:

(A) 16 **(B)** 14 **(C)** 12 **(D)** 8

Formula / idea: Number of functions from an m-element set to an n-element set $= n^m$.

1 Each of the 4 elements of A has 2 choices of image.
$\text{Total functions} = 2^4 = 16$

2 A function is not onto only when every element goes to the same image.
all to a or all to b : 2 functions

3 Subtract these.
$\text{Onto functions} = 16 - 2 = 14$ ✓ 1

4 Option (B).

Answer **(B)** 14

Watch out: Answering 16 — that counts all functions, not only the onto ones.

6 **Assertion-Reason** CBSE 2024 (65/4/1) Q19 1 mark

Assertion (A): The relation $R = \{(x, y) : (x + y)$ is a prime number and $x, y \in \mathbb{N}\}$ is not a reflexive relation. **Reason (R):** The number $2n$ is composite for all natural numbers n.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) **Assertion (A) is true, but Reason (R) is false.**

(D) Assertion (A) is false, but Reason (R) is true.

Formula / idea: $(x, x) \in R$ needs $x + x = 2x$ to be prime.

1	Check A: take $x = 2$. $2 + 2 = 4$ is not prime $\Rightarrow$ $(2, 2) \notin R$ so R is not reflexive: A is true	
2	Check R: take $n = 1$. $2n = 2,$ which is prime, not composite so R is false	✓ 1
3	A false statement cannot explain A.	
4	A is true, R is false. Option (C).	

Answer **(C) A is true, but R is false**

Watch out: Forgetting $n = 1$: then $2n = 2$ is prime, so R fails.

7 **Short answer** CBSE 2024 (65/1/1) Q26(b) 2 marks

A function f is defined from $\mathbb{R} \to \mathbb{R}$ as $f(x) = ax + b$, such that $f(1) = 1$ and $f(2) = 3$. Find function $f(x)$. Hence, check whether function $f(x)$ is one-one and onto or not.

Formula / idea: Onto: for every y in the codomain, solve $f(x) = y$ and check x lies in the domain.

1	Use the given values. $f(1) = a + b = 1, \qquad f(2) = 2a + b = 3$	
2	Subtract to get a, then find b. $a = 2,\ b = -1 \Rightarrow f(x) = 2x - 1$	✓ ½
3	One-one: let $f(x_1) = f(x_2)$. $2x_1 - 1 = 2x_2 - 1 \Rightarrow x_1 = x_2$	✓ ½

4	Onto: let $y \in \mathbb{R}$ and solve $f(x) = y$. $2x - 1 = y \Rightarrow x = \dfrac{y+1}{2} \in \mathbb{R}$	✓ ½
5	Check the pre-image. $f\left(\dfrac{y+1}{2}\right) = (y+1) - 1 = y$	✓ ½
6	Hence $f(x) = 2x - 1$ is both one-one and onto.	

Answer $f(x) = 2x - 1$**; it is both one-one and onto.**

Watch out: Finding $x = \dfrac{y+1}{2}$ but not showing $f(x) = y$ for it.

Level 2

19 marks

8 **Short answer** PYQ-style 2 marks

Show that the relation R on the set $\mathbb{Z}$ of integers, given by $R = \{(a, b) : 2a + 3b \text{ is divisible by } 5\}$, is an equivalence relation.

Formula / idea: An equivalence relation is reflexive, symmetric and transitive.

1	Reflexive: put $b = a$. $2a + 3a = 5a,$ divisible by $5 \Rightarrow (a, a) \in R$	✓ ½
2	Symmetric: let $(a, b) \in R$, so $2a + 3b = 5k$. $2b + 3a = 5(a + b) - (2a + 3b) = 5(a + b - k)$	✓ ½
3	So 5 divides $2b + 3a$, i.e. $(b, a) \in R$.	
4	Transitive: let $(a, b), (b, c) \in R$ and add the two conditions. $(2a + 3b) + (2b + 3c) = 5k + 5m \Rightarrow 2a + 5b + 3c = 5(k + m)$	
5	Rearrange. $2a + 3c = 5(k + m - b) \Rightarrow (a, c) \in R$	✓ ½
6	Hence R is an equivalence relation.	✓ ½

Answer R **is reflexive, symmetric and transitive, hence an equivalence relation.**

Watch out: Writing $2b + 3a = 2a + 3b$ — false in general; use $5(a + b)$ minus the given sum.

9 **Short answer** CBSE 2023 (65/1/1) Q32 3 marks

A relation R is defined on the set of real numbers $\mathbb{R}$ as $R = \{(x, y) : x \cdot y \text{ is an irrational number}\}$. Check whether R is reflexive, symmetric and transitive or not.

Formula / idea: To prove a property, argue for all elements; to disprove it, one counter-example is enough.

1	Reflexive? Take $x = 1$. $1 \cdot 1 = 1,\ \text{rational} \Rightarrow (1, 1) \notin R$ so R is not reflexive	✓ 1
2	Symmetric: let $(x, y) \in R$. $x \cdot y$ irrational $\Rightarrow$ $y \cdot x = x \cdot y$ irrational $\Rightarrow$ $(y, x) \in R$ multiplication of real numbers is commutative	✓ 1
3	Transitive? Take $x = 1, y = \sqrt{2}, z = 2$. $1 \cdot \sqrt{2} = \sqrt{2}, \quad \sqrt{2} \cdot 2 = 2\sqrt{2} \quad (\text{both irrational})$	
4	So $(1, \sqrt{2}) \in R$ and $(\sqrt{2}, 2) \in R$. But: $1 \cdot 2 = 2,\ \text{rational} \Rightarrow (1, 2) \notin R$	✓ 1
5	Hence R is symmetric, but neither reflexive nor transitive.	

Answer R **is symmetric, but neither reflexive nor transitive.**

Watch out: Trying to prove transitivity in general — pick a counter-example with a common middle term.

10 **Short answer** Board-pattern 3 marks

Let $A = \{1, 2, 3, \ldots, 9\}$ and R be the relation in $A \times A$ defined by $(a, b)\, R\, (c, d)$ if $a + b = c + d$ for $(a, b), (c, d) \in A \times A$. Prove that R is an equivalence relation. Also obtain the equivalence class $[(3, 5)]$.

Formula / idea: $[(a, b)]$ is the set of all pairs in $A \times A$ related to (a, b).

1	Reflexive: for any $(a, b) \in A \times A$. $a + b = a + b \Rightarrow (a, b)\, R\, (a, b)$	✓ ½
2	Symmetric: let $(a, b)\, R\, (c, d)$. $a + b = c + d \Rightarrow c + d = a + b \Rightarrow (c, d)\, R\, (a, b)$	✓ ½
3	Transitive: let $(a, b)\, R\, (c, d)$ and $(c, d)\, R\, (e, f)$. $a + b = c + d = e + f \Rightarrow (a, b)\, R\, (e, f)$	✓ ½
4	Hence R is an equivalence relation.	✓ ½
5	$[(3, 5)]$ contains every pair in $A \times A$ with sum $3 + 5 = 8$. $[(3, 5)] = \{(a, b) \in A \times A : a + b = 8\}$	

6	List them. $[(3,5)] = \{(1,7),(2,6),(3,5),(4,4),(5,3),(6,2),(7,1)\}$	✓ 1

Answer R **is an equivalence relation;** $[(3,5)] = \{(1,7),(2,6),(3,5),(4,4),(5,3),(6,2),(7,1)\}$

Watch out: Missing $(4,4)$, or including $(8,0)$ although $0 \notin A$.

11 **Short answer** PYQ-style 3 marks

Let $f : \mathbb{N} \to \mathbb{N}$ be defined by $f(n) = \begin{cases} n+1, & \text{if } n \text{ is odd} \\ n-1, & \text{if } n \text{ is even} \end{cases}$. Show that f is both one-one and onto.

Formula / idea: For a piecewise function, check one-one in every case, including mixed cases.

1	One-one: let $f(n_1) = f(n_2)$. Case 1: both odd, or both even. $n_1 + 1 = n_2 + 1$ or $n_1 - 1 = n_2 - 1 \Rightarrow n_1 = n_2$	✓ ½
2	Case 2: one odd, one even. Say n_1 odd, n_2 even. $n_1 + 1 = n_2 - 1 \Rightarrow n_2 = n_1 + 2$	
3	But $n_1 + 2$ is odd and n_2 is even — impossible.	✓ ½
4	The case n_1 even, n_2 odd is the same. So f is one-one.	✓ ½
5	Onto: let $m \in \mathbb{N}$. If m is odd, $m+1$ is even. $f(m+1) = (m+1) - 1 = m$	✓ ½
6	If m is even, $m-1$ (≥ 1) is odd. $f(m-1) = (m-1) + 1 = m$	✓ ½
7	Every $m \in \mathbb{N}$ has a pre-image. Hence f is one-one and onto.	✓ ½

Answer f **is both one-one and onto (bijective).**

Watch out: Skipping the mixed case (n_1 odd, n_2 even) in the one-one part.

12 **Short answer** CBSE 2024 (65/2/3) Q33(a) 3 marks

Show that a function $f : \mathbb{R} \to \mathbb{R}$ defined as $f(x) = x^2 + x + 1$ is neither one-one nor onto. Also, find all the values of x for which $f(x) = 3$.

Formula / idea: Two different inputs with the same output ⇒ not one-one; a codomain value never reached ⇒ not onto.

1	Solve $f(x) = 3$. $x^2 + x + 1 = 3 \Rightarrow x^2 + x - 2 = 0$	

2	Factorise. $(x+2)(x-1)=0 \Rightarrow x=1$ or $x=-2$	✓ 1
3	So $f(1)=f(-2)=3$, but $1 \neq -2$. $\Rightarrow$ f is not one-one	✓ 1
4	Complete the square. $f(x)=\left(x+\frac{1}{2}\right)^2+\frac{3}{4} \geq \frac{3}{4}$	
5	So $0 \in \mathbb{R}$ (codomain) has no pre-image: f is not onto. $f(x) \neq 0$ for every $x \in \mathbb{R}$	✓ 1
6	Hence f is neither one-one nor onto, and $f(x)=3$ at $x=1,-2$.	

Answer **Neither one-one nor onto; $f(x)=3$ for $x=1$ and $x=-2$.**

Watch out: Saying 'not onto' without naming a codomain value (like 0) that has no pre-image.

13 **Short answer** CBSE 2023 (65/4/1) Q34(b) 3 marks

Let $f: \mathbb{R}-\left\{-\frac{4}{3}\right\} \to \mathbb{R}$ be a function defined as $f(x)=\frac{4x}{3x+4}$. Show that f is a one-one function. Also, check whether f is an onto function or not.

Formula / idea: Onto fails if some y in the codomain gives no valid x when solving $f(x)=y$.

1	One-one: let $f(x_1)=f(x_2)$ and cross-multiply. $4x_1(3x_2+4)=4x_2(3x_1+4)$	✓ ½
2	Expand and cancel $12x_1x_2$. $16x_1=16x_2 \Rightarrow x_1=x_2$	✓ 1
3	So f is one-one.	
4	Onto: let $y \in \mathbb{R}$ and solve $f(x)=y$. $4x=3xy+4y \Rightarrow x=\frac{4y}{4-3y}$, $y \neq \frac{4}{3}$	✓ ½
5	Check $y=\frac{4}{3}$ directly. $\frac{4x}{3x+4}=\frac{4}{3} \Rightarrow 12x=12x+16 \Rightarrow 0=16$	✓ ½
6	So $\frac{4}{3} \in \mathbb{R}$ has no pre-image. Hence f is one-one but not onto.	✓ ½

Answer **f is one-one but not onto ($\frac{4}{3}$ has no pre-image).**

Watch out: Calling it onto — the codomain is all of $\mathbb{R}$, and $\frac{4}{3}$ is never reached.

14 **Short answer** CBSE 2024 (65/1/1) Q26(a) 2 marks

A relation R on set $A = \{1, 2, 3, 4, 5\}$ is defined as $R = \{(x, y) : |x^2 - y^2| < 8\}$. Check whether the relation R is reflexive, symmetric and transitive.

Formula / idea: Write the relation in roster form, then test each property.

1	Write R in roster form. $R = \{(1,1), (2,2), (3,3), (4,4), (5,5),$ $(1,2), (2,1), (2,3), (3,2), (3,4), (4,3)\}$	
2	Reflexive: for every $x \in A$. $\lvert x^2 - x^2 \rvert = 0 < 8 \Rightarrow (x, x) \in R$	✓ ½
3	Symmetric: let $(x, y) \in R$. $\lvert y^2 - x^2 \rvert = \lvert x^2 - y^2 \rvert < 8 \Rightarrow (y, x) \in R$	✓ ½
4	Transitive? $(1, 2) \in R$ and $(2, 3) \in R$, but: $\lvert 1^2 - 3^2 \rvert = 8 \not< 8 \Rightarrow (1, 3) \notin R$	✓ 1
5	Hence R is reflexive and symmetric, but not transitive.	

Answer R **is reflexive and symmetric, but not transitive.**

Watch out: Treating $8 < 8$ as true — the condition is strict, so $(1, 3) \notin R$.

Level 3

25 marks

15 **Short answer** NCERT Ex 1.1 Q12 3 marks

Show that the relation R defined in the set A of all triangles as $R = \{(T_1, T_2) : T_1 \text{ is similar to } T_2\}$ is an equivalence relation. Consider three right angle triangles T_1 with sides $3, 4, 5$, T_2 with sides $5, 12, 13$ and T_3 with sides $6, 8, 10$. Which triangles among T_1, T_2 and T_3 are related?

Formula / idea: Two triangles are similar when their corresponding sides are proportional.

1	Reflexive: every triangle is similar to itself. $(X, X) \in R$ for every triangle X	✓ ½
2	Symmetric: if X is similar to Y, then Y is similar to X. $(X, Y) \in R \Rightarrow (Y, X) \in R$	✓ ½
3	Transitive: if $X \sim Y$ and $Y \sim Z$, then $X \sim Z$. $(X, Y), (Y, Z) \in R \Rightarrow (X, Z) \in R$	✓ ½
4	Hence R is an equivalence relation.	✓ ½

5	Compare the sides of T_1 and T_3. $\frac{3}{6} = \frac{4}{8} = \frac{5}{10} = \frac{1}{2} \Rightarrow T_1 \sim T_3$	✓ ½
6	Compare T_2 with T_1 and with T_3. $\frac{3}{5} \neq \frac{4}{12}, \quad \frac{6}{5} \neq \frac{8}{12}$	✓ ½
7	Hence only T_1 and T_3 are related.	

Answer R **is an equivalence relation; only** T_1 **and** T_3 **are related.**

Watch out: Saying all right triangles are similar — a right angle alone does not make sides proportional.

16 **Long answer** CBSE 2023 (65/4/1) Q34(a) 5 marks

If $\mathbb{N}$ denotes the set of all natural numbers and R is the relation on $\mathbb{N} \times \mathbb{N}$ defined by $(a, b)\, R\, (c, d)$ if $ad(b + c) = bc(a + d)$, show that R is an equivalence relation.

Formula / idea: Dividing by $abcd$ turns the condition into $\frac{1}{a} - \frac{1}{b} = \frac{1}{c} - \frac{1}{d}$.

1	Divide both sides by $abcd$ (non-zero, all are natural numbers). $\frac{b+c}{bc} = \frac{a+d}{ad} \Rightarrow \frac{1}{c} + \frac{1}{b} = \frac{1}{d} + \frac{1}{a}$	
2	Rearrange into a handy form. $(a, b)\, R\, (c, d) \iff \frac{1}{a} - \frac{1}{b} = \frac{1}{c} - \frac{1}{d}$	✓ 1
3	Reflexive: for any $(a, b) \in \mathbb{N} \times \mathbb{N}$. $ab(b + a) = ba(a + b) \Rightarrow (a, b)\, R\, (a, b)$	✓ 1
4	Symmetric: let $(a, b)\, R\, (c, d)$ and rewrite its equation. $ad(b + c) = bc(a + d) \Rightarrow cb(d + a) = da(c + b)$	✓ ½
5	This is the condition for $(c, d)\, R\, (a, b)$. So R is symmetric.	✓ ½
6	Transitive: let $(a, b)\, R\, (c, d)$ and $(c, d)\, R\, (e, f)$. $\frac{1}{a} - \frac{1}{b} = \frac{1}{c} - \frac{1}{d}$ and $\frac{1}{c} - \frac{1}{d} = \frac{1}{e} - \frac{1}{f}$	
7	Equate the two results. $\frac{1}{a} - \frac{1}{b} = \frac{1}{e} - \frac{1}{f}$	✓ ½
8	Multiply by $abef$ and rearrange. $bef - aef = abf - abe \Rightarrow af(b + e) = be(a + f)$	✓ 1

9	So $(a,b)\,R\,(e,f)$: R is transitive. Hence R is an equivalence relation.	✓ ½

Answer R **is reflexive, symmetric and transitive, hence an equivalence relation.**

Watch out: Attempting transitivity directly from $ad(b+c)=bc(a+d)$ — use the reciprocal form first.

17 **Long answer** Board-pattern 5 marks

Show that the function $f:\mathbb{R}\to\mathbb{R}$ defined by $f(x)=x|x|$ is both one-one and onto.

Formula / idea: $|x|=x$ for $x\geq 0$ and $|x|=-x$ for $x<0$.

1	Remove the modulus. $f(x)=\begin{cases}x^2, & x\geq 0\\ -x^2, & x<0\end{cases}$	✓ ½
2	Note: $f(x)$ has the same sign as x. $x\geq 0\Rightarrow f(x)\geq 0,\qquad x<0\Rightarrow f(x)<0$	
3	One-one: let $f(x_1)=f(x_2)$. Case 1: $x_1,x_2\geq 0$. $x_1^2=x_2^2\ \Rightarrow\ x_1=x_2$ both are non-negative	✓ ½
4	Case 2: $x_1,x_2<0$. $-x_1^2=-x_2^2\ \Rightarrow\ x_1=x_2$ both are negative	✓ ½
5	Case 3: $x_1\geq 0$ and $x_2<0$. $f(x_1)\geq 0>f(x_2)\ \Rightarrow\ f(x_1)\neq f(x_2)$	✓ ½
6	Same if $x_1<0\leq x_2$. So case 3 cannot occur; f is one-one.	✓ ½
7	Onto: let $y\in\mathbb{R}$. If $y\geq 0$, take $x=\sqrt{y}$. $f(\sqrt{y})=\sqrt{y}\cdot\sqrt{y}=y$	✓ 1
8	If $y<0$, take $x=-\sqrt{-y}$, a negative number. $f(-\sqrt{-y})=-\left(\sqrt{-y}\right)^2=-(-y)=y$	✓ 1
9	Every $y\in\mathbb{R}$ has a pre-image. Hence f is one-one and onto.	✓ ½

Answer f **is both one-one and onto (bijective).**

Watch out: Using $x=\sqrt{y}$ for negative y — it is undefined; take $x=-\sqrt{-y}$.

18 **Long answer** CBSE 2024 (65/3/1) Q32 5 marks

A relation R on set $A = \{-4, -3, -2, -1, 0, 1, 2, 3, 4\}$ is defined as $R = \{(x, y) : x + y \text{ is an integer divisible by } 2\}$. Show that R is an equivalence relation. Also, write the equivalence class $[2]$.

Formula / idea: $[a] = \{x \in A : (x, a) \in R\}$.

1	Reflexive: for every $x \in A$. $x + x = 2x,$ divisible by $2 \Rightarrow (x, x) \in R$	✓ 1
2	Symmetric: let $(x, y) \in R$, so $x + y = 2k$. $y + x = x + y = 2k \Rightarrow (y, x) \in R$	✓ 1
3	Transitive: let $(x, y), (y, z) \in R$ and add the two conditions. $(x + y) + (y + z) = 2k + 2m \Rightarrow x + 2y + z = 2(k + m)$	
4	Rearrange. $x + z = 2(k + m - y) \Rightarrow (x, z) \in R$	✓ 1½
5	Hence R is an equivalence relation.	✓ ½
6	$[2]$ has all $y \in A$ with $2 + y$ even, i.e. y even. $[2] = \{y \in A : 2 + y \text{ is divisible by } 2\}$	
7	List them. $[2] = \{-4, -2, 0, 2, 4\}$	✓ 1

Answer R **is an equivalence relation;** $[2] = \{-4, -2, 0, 2, 4\}$

Watch out: Leaving out 0 or the negative even numbers from $[2]$.

19 **Short answer** CBSE 2023 (65/2/2) Q34 3 marks

A function $f : [-4, 4] \to [0, 4]$ is given by $f(x) = \sqrt{16 - x^2}$. Show that f is an onto function but not a one-one function. Further, find all possible values of a for which $f(a) = \sqrt{7}$.

Formula / idea: Onto: every y in the codomain $[0, 4]$ must have a pre-image in the domain $[-4, 4]$.

1	Onto: let $y \in [0, 4]$ and solve $f(x) = y$. $\sqrt{16 - x^2} = y \Rightarrow x^2 = 16 - y^2 \Rightarrow x = \pm\sqrt{16 - y^2}$	
2	As $0 \le y \le 4$, take $x = \sqrt{16 - y^2} \in [0, 4] \subset [-4, 4]$. $f(x) = \sqrt{16 - (16 - y^2)} = \sqrt{y^2} = y$ $y \ge 0$, so $\sqrt{y^2} = y$	✓ 1
3	Every $y \in [0, 4]$ has a pre-image, so f is onto.	

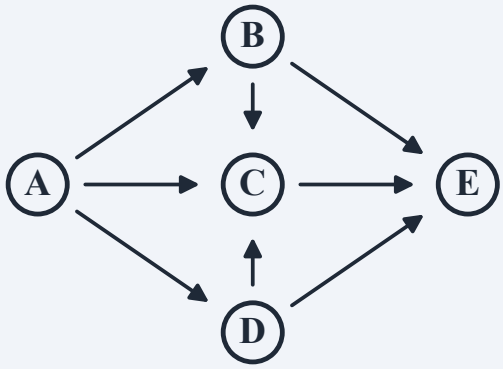
4	Not one-one: take $x = 4$ and $x = -4$. $f(4) = f(-4) = \sqrt{16 - 16} = 0,\ \text{but } 4 \neq -4$	✓ 1
5	Put $f(a) = \sqrt{7}$ and square. $16 - a^2 = 7 \ \Rightarrow\ a^2 = 9 \ \Rightarrow\ a = \pm 3$	✓ ½
6	Both lie in $[-4, 4]$. So $a = 3$ or $a = -3$.	✓ ½

Answer f **is onto but not one-one;** $a = 3$ **or** $a = -3$**.**

Watch out: Giving only $a = 3$ — the negative value -3 is also in the domain.

20 **Case study** CBSE SQP 2025-26 Q36 4 marks

Case Study: A city's traffic management department is planning to optimize traffic flow by analyzing the connectivity between various traffic signals. The city has five major spots labelled A, B, C, D and E. The department has collected the following data regarding one-way traffic flow between spots: traffic flows from A to B, A to C and A to D; from B to C and B to E; from C to E; from D to E and D to C. The department wants to represent and analyze this data using relations and functions, taking $(X, Y) \in R$ when traffic flows from X to Y, on the set $S = \{A, B, C, D, E\}$.



arrow $X \to Y$: one-way traffic from X to Y

(i) Is the traffic flow reflexive? Justify. 1 mark

(ii) Is the traffic flow transitive? Justify. 1 mark

(iii)(a) Represent the relation describing the traffic flow as a set of ordered pairs. Also state the domain and range of the relation. 2 marks

OR

(iii)(b) Does the traffic flow represent a function? Justify your answer. 2 marks

Formula / idea: Domain of a relation = set of first entries; range = set of second entries.

(i) 1 mark

1	Reflexive needs $(X, X) \in R$ for every spot X.	
2	No traffic flows from a spot to itself. $(A, A) \notin R$	✓ 1

3	Hence the traffic flow is not reflexive.	

Answer (i) **Not reflexive, since $(A, A) \notin R$.**

(ii)

1 mark

1	Traffic flows from A to B and from B to E. $(A, B) \in R,\ (B, E) \in R$	
2	But there is no direct flow from A to E. $(A, E) \notin R$	✓ 1
3	Hence the traffic flow is not transitive.	

Answer (ii) **Not transitive, since $(A, B), (B, E) \in R$ but $(A, E) \notin R$.**

(iii)(a)

2 marks

1	Write each one-way flow as an ordered pair. $R = \{(A, B), (A, C), (A, D), (B, C),$ $(B, E), (C, E), (D, E), (D, C)\}$	✓ 1
2	Domain = set of first entries. $\text{Domain} = \{A, B, C, D\}$	✓ ½
3	Range = set of second entries. $\text{Range} = \{B, C, D, E\}$	✓ ½

Answer (iii)(a) **R has 8 pairs; domain $\{A, B, C, D\}$, range $\{B, C, D, E\}$.**

OR

(iii)(b)

2 marks

1	A function needs exactly one image for every element of S.	
2	A is related to three different spots. $(A, B),\ (A, C),\ (A, D) \in R$	✓ 1
3	Also E has no image at all. no pair $(E, Y) \in R$	✓ ½
4	Hence the traffic flow does not represent a function.	✓ ½

Answer (iii)(b) **No — A has three images and E has none.**

Watch out: Writing the range as all five spots — A never appears as a second entry.